Engineering Specification

Project : Epsilon Data Center

Vendor : ABB

Equipment : Pump

Issue Date : 2025-04-27

Project Overview

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Cooling System

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Supply air temperature shall remain between 18°C and 27°C.

Each cooling unit shall include vibration monitoring sensors.

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Cooling redundancy shall be verified during commissioning.

Cooling equipment shall support SNMP monitoring.

Chilled water pumps shall operate with redundant motors.

CRAH units shall support automatic failover.

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Electrical System

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Power Distribution Units shall provide branch circuit monitoring.

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Electrical panels shall comply with IEC standards.

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UPS capacity shall be 500 KVA.

Electrical panels shall comply with IEC standards.

Generator backup shall support 72 hours of continuous operation.

Power Distribution Units shall provide branch circuit monitoring.

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Emergency shutdown shall disconnect non-critical loads.

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Fire Protection

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Smoke detectors shall be installed in all electrical rooms.

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FM200 suppression shall protect all server rooms.

Fire alarm panels shall integrate with the BMS.

Emergency evacuation procedures shall be displayed.

VESDA detection shall be installed.

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Network Infrastructure

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Core switches shall support redundant uplinks.

All network racks shall use Cat6A cabling.

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Network monitoring shall integrate with the NOC.

Fiber optic backbone shall use single-mode fiber.

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Generator synchronization tests shall be completed.

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Integrated System Testing shall be completed before handover.

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Commissioning reports shall include all test results.

Cooling performance shall be verified under full load.

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Quality Assurance

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Inspection reports shall be approved before the next activity.

All materials shall comply with project specifications.

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Vendor documentation shall be reviewed before installation.

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